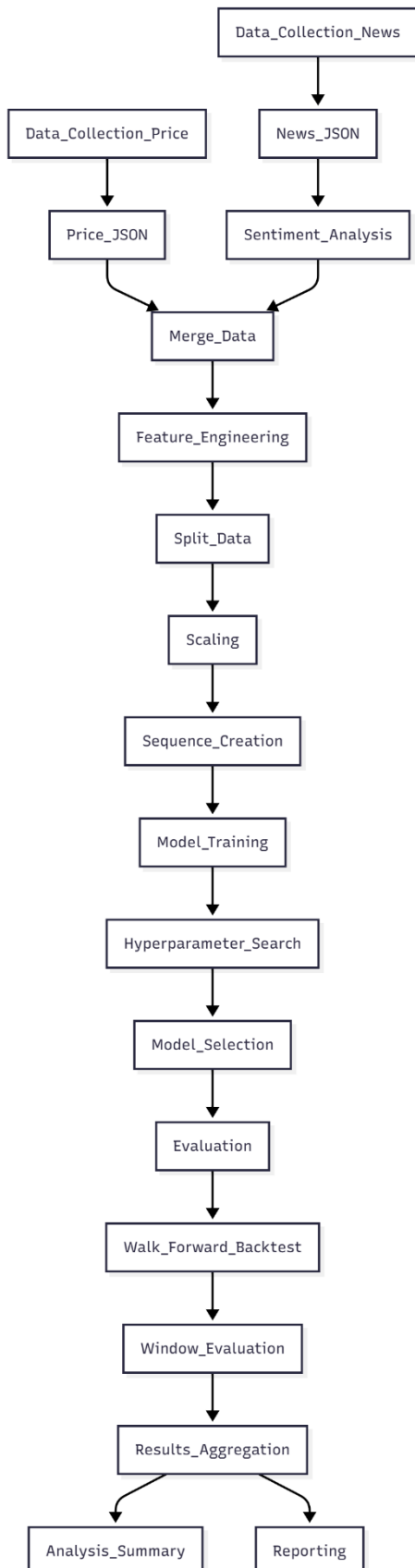


PHASE-3 DLT PROJECT



WORKING OF MODEL FINBERT LSTM

1) Data Collection: Gather historical price data (using yfinance) and news data (using Alpaca News API) for selected financial symbols (AAPL, TSLA, MSFT, GOOGL, NVDA). Run sentiment analysis on news using FinBERT.

2) Data Preprocessing: Merge price, sentiment, and other features. Engineer features such as OHLCV and averaged sentiment scores per day. Split data temporally for training, validation, and testing periods.

3) Model Training: Scale features, create time-ordered LSTM sequences, train FinBERT-LSTM and optimize via hyperparameter search (layers, hidden units, etc.).

4) Backtesting: Run walk-forward validation, aggregating out-of-sample results for each moving window (no data leakage).

5) Reporting & Analysis: Evaluate returns, win rates, and error metrics, and generate performance plots and summary reports.

GOALS AND ERROR METRICS

Goals:

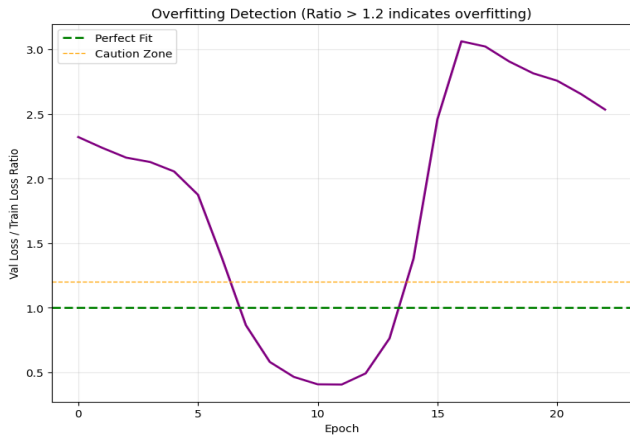
- Predict next-day closing price (regression) or directional change (classification) for stocks.
- Construct a trading/buy-sell signal based on price prediction or sentiment score.

Error Metrics:

- **Regression:** Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE).
- **Directional Prediction:** Direction accuracy (percentage of days where predicted up/down matches actual direction).
- **Financial Performance:** Portfolio return, Sharpe-like ratio, win rate.
- **Target Values:**
 - Example targets: $RMSE < 2.5$, Direction accuracy $> 55\%$, Positive portfolio return (benchmark $> 0\%$).

OPTIMISATION CURVES





TRAINING SUMMARY

Total Epochs: 23

Final Training Loss: 11889.342773

Final Validation Loss: 30121.968750

Best Validation Loss: 13177.222656

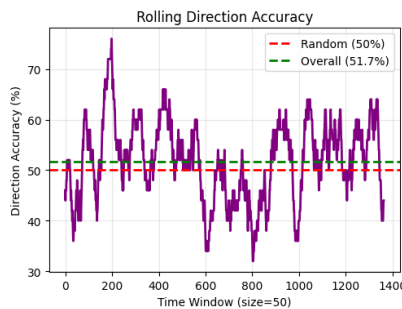
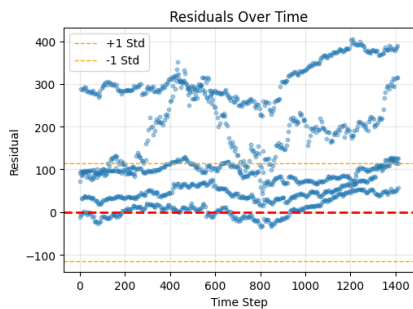
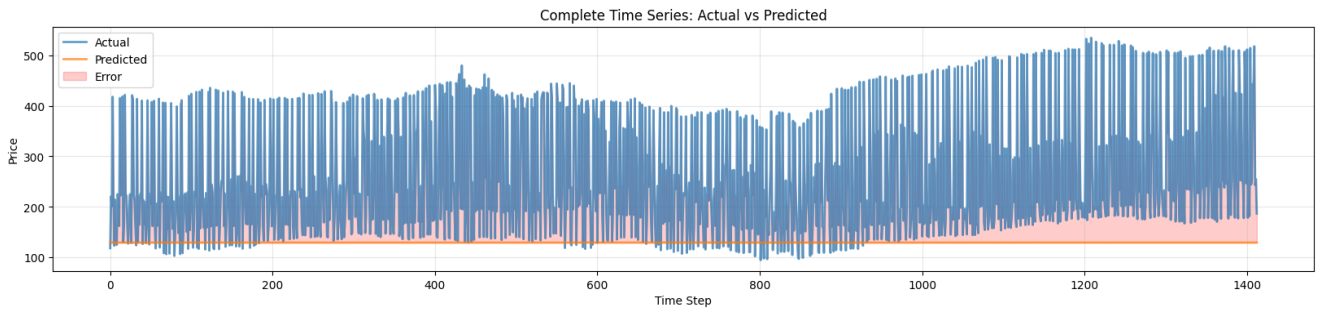
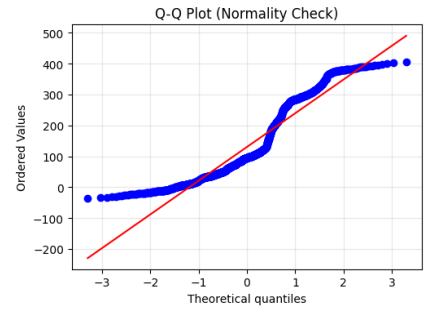
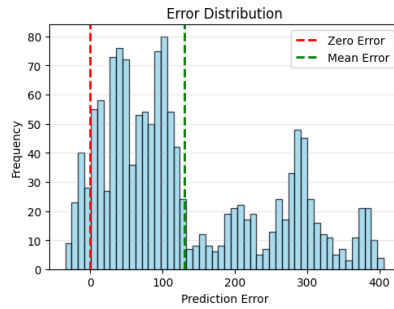
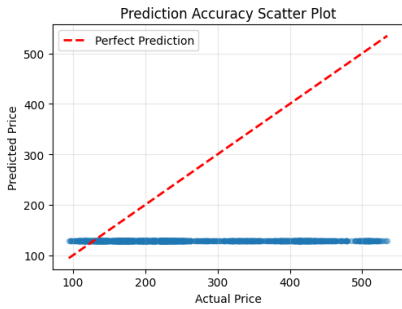
Best Epoch: 7

Overfitting Ratio: 2.534

Status: Overfitting

Model Architecture:

- Hidden Size: 32
- Num Layers: 1
- Dropout: 0.2
- Attention: True
- Bidirectional: False
- Learning Rate: 0.1



PERFORMANCE METRICS SUMMARY

Regression Metrics:

- MSE: 30121.970748
- MAE: 132.134004
- RMSE: 173.556823

Statistical Metrics:

- Mean Error: 130.231150
- Std Error: 114.724968
- Min Error: -34.696571
- Max Error: 405.717438

Direction Prediction:

- Accuracy: 51.70%
- Up Correct: 366
- Down Correct: 364

Sample Statistics:

- Total Samples: 1413
- Validation Period: 1413 days

MODEL SUMMARY

MODEL ARCHITECTURE

Model Type:	FinBERT-LSTM Advanced
Input Size:	10 features
Hidden Size:	32 units
Number of Layers:	1
Dropout Rate:	0.2
Attention Mechanism:	Enabled
Bidirectional:	No
Attention Heads:	8
Attention Type:	Multi-Head Self-Attention
LSTM Output Size:	32
FC1 Hidden Size:	16
FC2 Output Size:	1 (price prediction)

TRAINING CONFIGURATION

Learning Rate:	0.1
Sequence Length:	30 days

Group Members:

- 1)Shikhar Uniyal-230968298 (roll no:43)
- 2)Aadarsh Prasad-230968280(roll no:39)
- 3)Chaitanya Kad-230968286(roll no:40)

